

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import urllib.request
```

```
img = cv2.imread("image.png", 0)

plt.figure(figsize=(6,6))
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis("off")
plt.show()
```

Original Image



```
f = np.fft.fft2(img)

fshift = np.fft.fftshift(f)

magnitude_spectrum = 20 * np.log(np.abs(fshift) + 1)
```

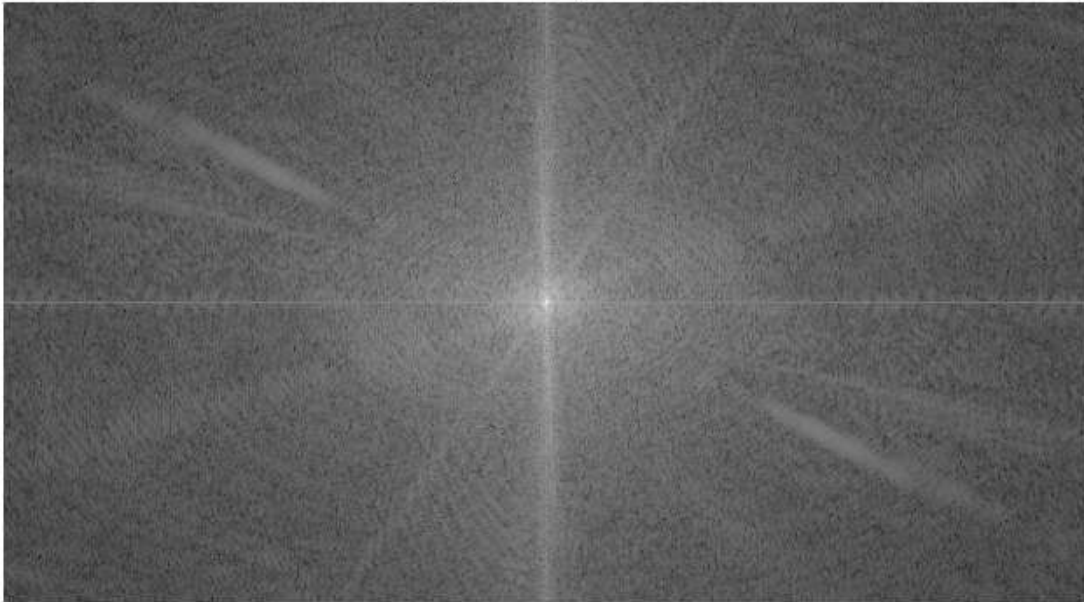
```
plt.figure(figsize=(8,6))

plt.imshow(magnitude_spectrum,
           cmap='gray')

plt.title("Magnitude Spectrum")
plt.axis("off")

plt.show()
```

Magnitude Spectrum



```
rows, cols = img.shape

crow, ccol = rows//2, cols//2

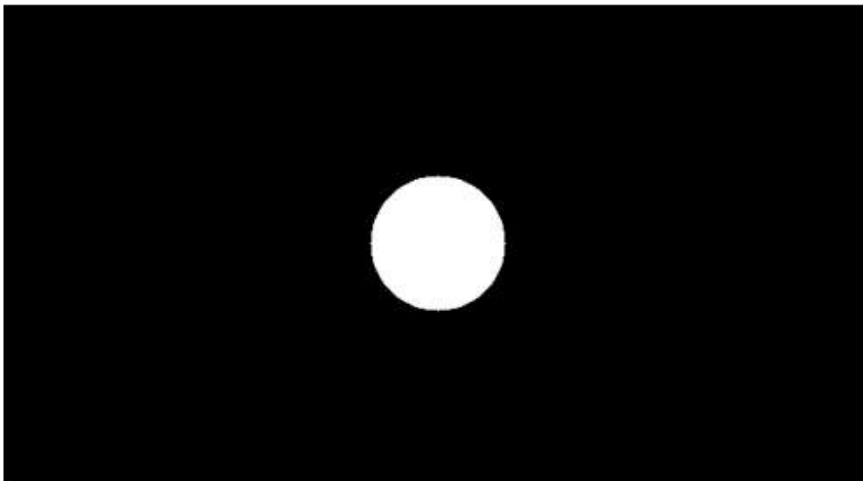
mask_lp = np.zeros((rows, cols),
                    np.uint8)

r = 50

cv2.circle(mask_lp,
            (ccol,crow),
            r,
            1,
            -1)

plt.imshow(mask_lp,cmap='gray')
plt.title("Low Pass Filter Mask")
plt.axis("off")
plt.show()
```

Low Pass Filter Mask



```
fshift_lp = fshift * mask_lp
```

```
img_lp = np.fft.ifft2(  
    np.fft.ifftshift(fshift_lp)  
)  
  
img_lp = np.abs(img_lp)
```

```
plt.figure(figsize=(6,6))  
  
plt.imshow(img_lp,  
           cmap='gray')  
  
plt.title("Low Pass Filtered Image")  
plt.axis("off")  
plt.show()
```

Low Pass Filtered Image



```
mask_hp = np.ones((rows, cols),  
                  np.uint8)  
  
cv2.circle(mask_hp,  
           (ccol,crow),  
           r,  
           0,  
           -1)  
  
plt.imshow(mask_hp,  
           cmap='gray')  
  
plt.title("High Pass Filter Mask")  
plt.axis("off")  
plt.show()
```

High Pass Filter Mask



```
fshift_hp = fshift * mask_hp

img_hp = np.fft.ifft2(
    np.fft.ifftshift(fshift_hp)
)

img_hp = np.abs(img_hp)
```

```
plt.figure(figsize=(6,6))

plt.imshow(img_hp,
           cmap='gray')

plt.title("High Pass Filtered Image")
plt.axis("off")
plt.show()
```

High Pass Filtered Image



```
plt.figure(figsize=(15,10))

plt.subplot(2,2,1)
plt.imshow(img,cmap='gray')
plt.title("Original")
plt.axis("off")

plt.subplot(2,2,2)
plt.imshow(magnitude_spectrum,
```

```

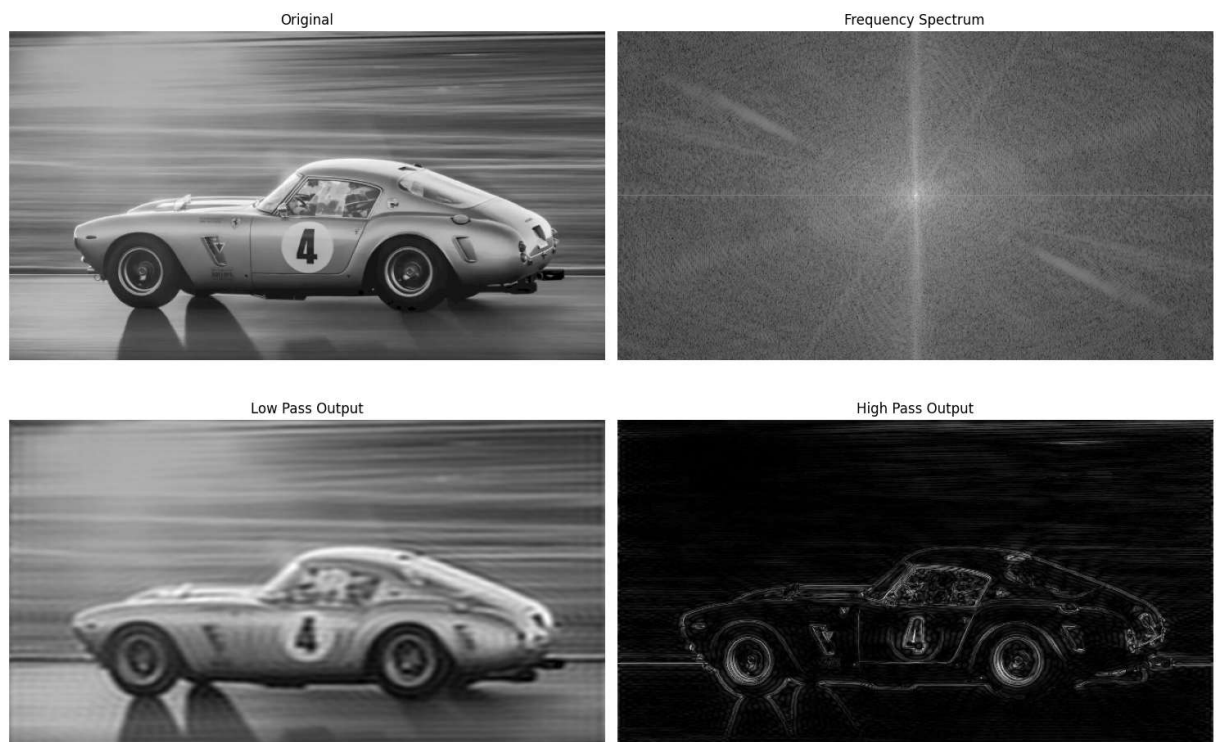
cmap= 'gray')
plt.title("Frequency Spectrum")
plt.axis("off")

plt.subplot(2,2,3)
plt.imshow(img_lp,cmap='gray')
plt.title("Low Pass Output")
plt.axis("off")

plt.subplot(2,2,4)
plt.imshow(img_hp,cmap='gray')
plt.title("High Pass Output")
plt.axis("off")

plt.tight_layout()
plt.show()

```



```

print("""
Observations:

```

1. Fourier Transform converts image into frequency domain.
 2. Low Pass Filter removes high-frequency components.
 3. Low Pass output appears smoother and less noisy.
 4. High Pass Filter removes low-frequency components.
 5. High Pass output highlights edges and fine details.
 6. Frequency filtering is effective for image enhancement.
- ```

""")

```

Observations:

1. Fourier Transform converts image into frequency domain.
2. Low Pass Filter removes high-frequency components.
3. Low Pass output appears smoother and less noisy.
4. High Pass Filter removes low-frequency components.
5. High Pass output highlights edges and fine details.
6. Frequency filtering is effective for image enhancement.

```
print("""
Result:

Frequency domain image filtering was successfully
implemented using Fourier Transform.

Low Pass Filtering reduced noise and smoothed
the image, whereas High Pass Filtering enhanced
edges and important image details.
""")
```

Result: